



Velkommen til Netværk II

Jeg hedder Grete Teta, jeg har undervist på skolen siden 2020

Jeg er mest på H3 svendeprøveforløbet, men har også haft H1, H2 og H3 anden del.

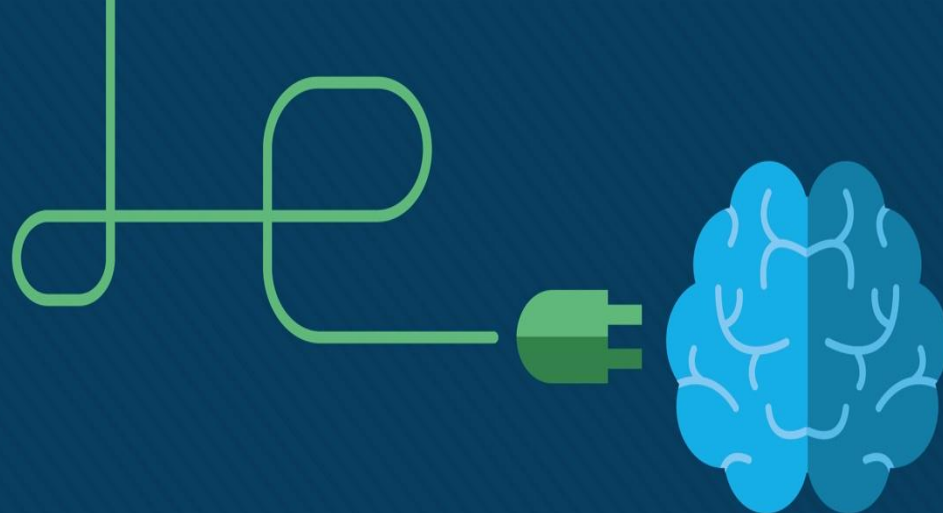
Jeg laver mange af opgaverne selv, vi kommer også til at konstruere nogle opgaver på tavlen, så I får repeteret Netværk I

Jeg bruger også netacad opgaver ind imellem.

Der er SKAL opgaver, som I bør aflevere, hvis I vil bestå faget.

Der vil også være ekstra opgaver til de hurtige, så de ikke sidder å keder sig.

Har I haft om FHRP og HSRP?



Module 1: Single-Area OSPFv2 Concepts

Module 2: Single-Area OSPFv2 Configuration





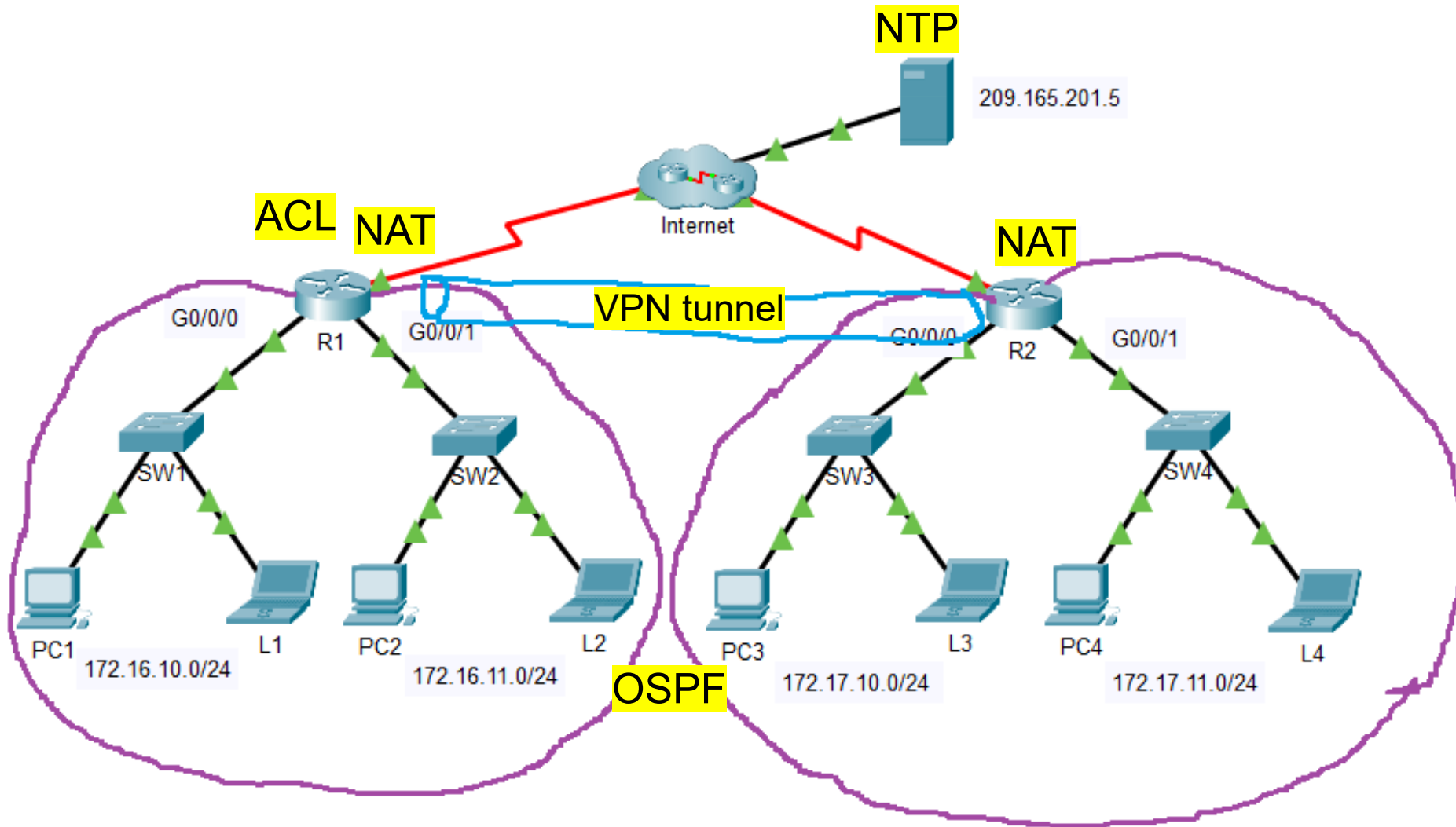
Indhold i modul 1 og 2

- Lidt om statisk vs. dynamisk routing (som I jo har hørt om i Netværk1 😊)
- Generel snak om OSPF
- Hvordan konfigurerer man OSPF
- Fejlsøgning i OSPF

(ca 30 dias + video)



Hvad kan jeg efter Netværk 2?



Samt "kendskab til" Design og Management; IPv6; QoS; WAN samt Virtualisering; Automatisering; Security og TSHOOT



Plan – som den ser ud lige nu 😊

Uge/dag	Mandag	Tirsdag	Onsdag	Torsdag	Fredag
Uge 37 7/9 – 11/9 	Netværk II 08:15 – 13:35 6 lektioner Velkommen til Netværk II Module 1 Single-Area OSPF Module 2 Single-Area OSPF Configuration	Netværk II 08:15 – 15:15 8 lektioner Module 4 ACL Concepts Module 5 ACLs for IPv4 Configuration Module 6 NAT for IPv4	Netværk II 08:15 – 15:15 8 lektioner Module 3 Network Security Concepts Module 7 WAN Concepts	Netværk II 08:15 – 15:30 8 lektioner Module 8 VPN and IPsec Concepts Module 9 QoS Concepts	Netværk II 08:15 – 13:15 6 lektioner Module 10 Network Management <u>Hjemmestrikket opgave</u>
Uge 38 14/9 – 18/9	Netværk II 08:15 – 13:15 6 lektioner Module 11 Network Design Module 12 Network <u>Tshoot</u>	Netværk II 08:15 – 15:35 8 lektioner Module 13 Network Virtualization Module 14 Network Automation	Netværk II 08:15 – 15:30 8 lektioner Repetitionsopgave	Netværk II 08:15 – 15:15 8 lektioner Afsluttende Opgave løses individuelt Evaluering og opfølgning på faget og undervisning mv	

Planen findes naturligvis stadig i Learn – nu under Netværk 2



Vi starter blødt ud 😊

- Der er selvfølgelig masser af videoer derude.....
- Denne her (video) er ældgammel, men som opsamling og forståelse for hvordan en pakke/frame kan bevæge sig fra afsender til modtager, er den helt fin. Og nogle af jer kunne godt bruge den opsamling 😊

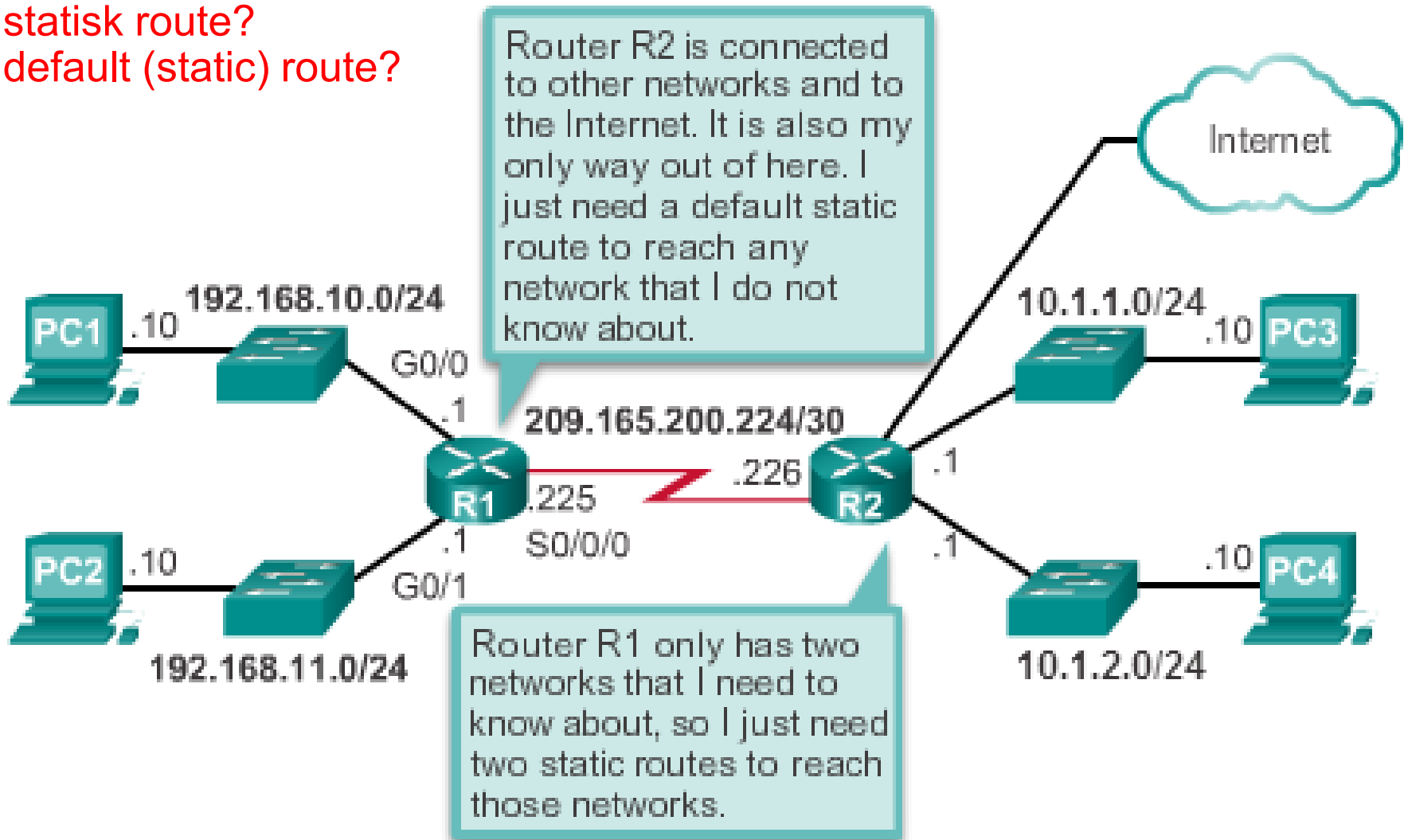
(den er lidt lang..... 16 minutter!)

- Efter videoen skal du endelig spørge hvis der er noget der ikke lige var klart, eller du ikke forstår 😊 Det skal du altid 😊
- [Packet Walkthrough: A PC Perspective \(youtube.com\)](#)

Static Routing

Static and Default Route Scenario

- Hvornår bruger man:
- statisk route?
 - default (static) route?

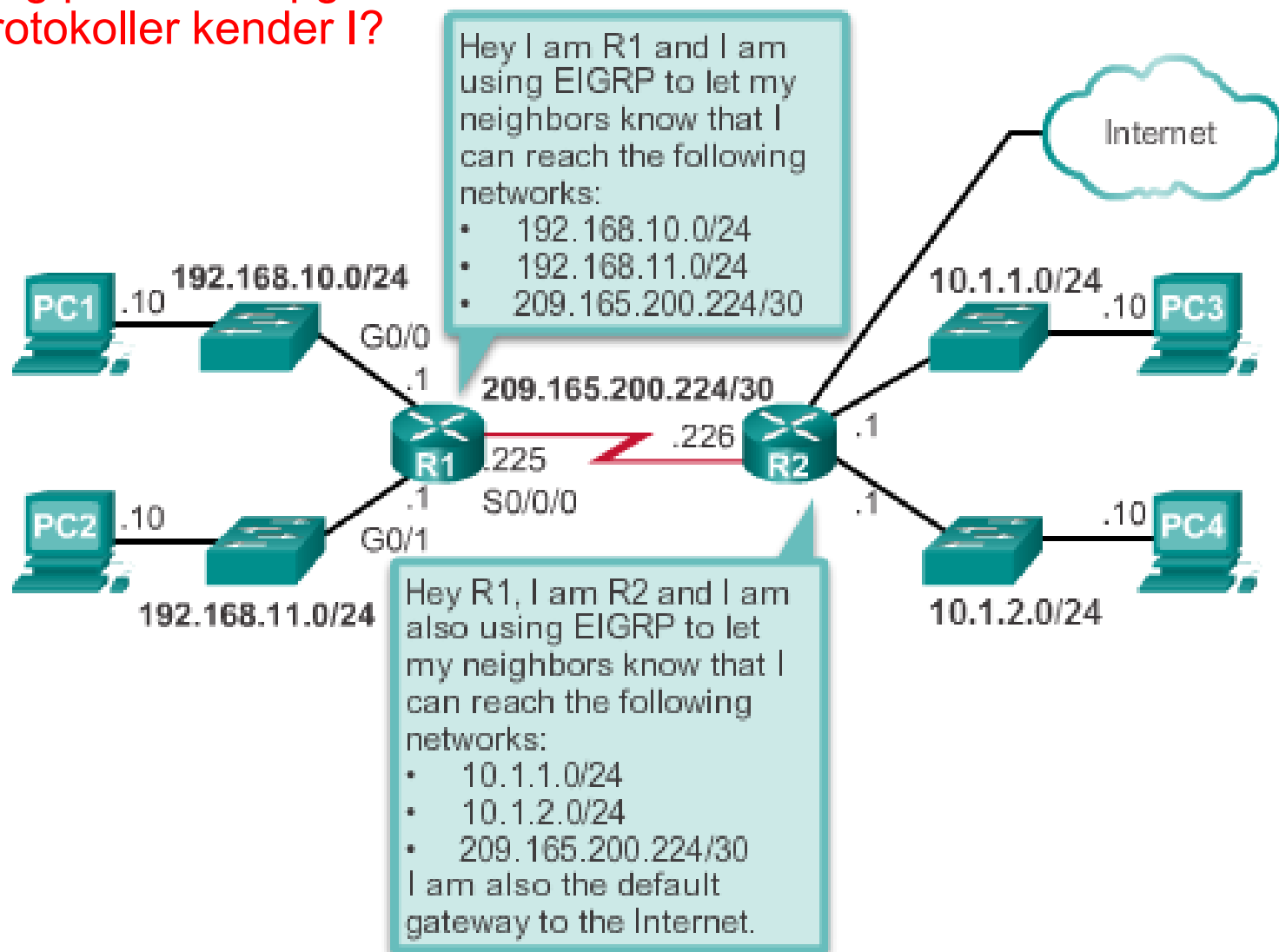


Dynamic Routing Protocols

Dynamic Routing Protocol Scenario

Hvad er en routing protocols opgaver?

Hvilke routing protokoller kender I?

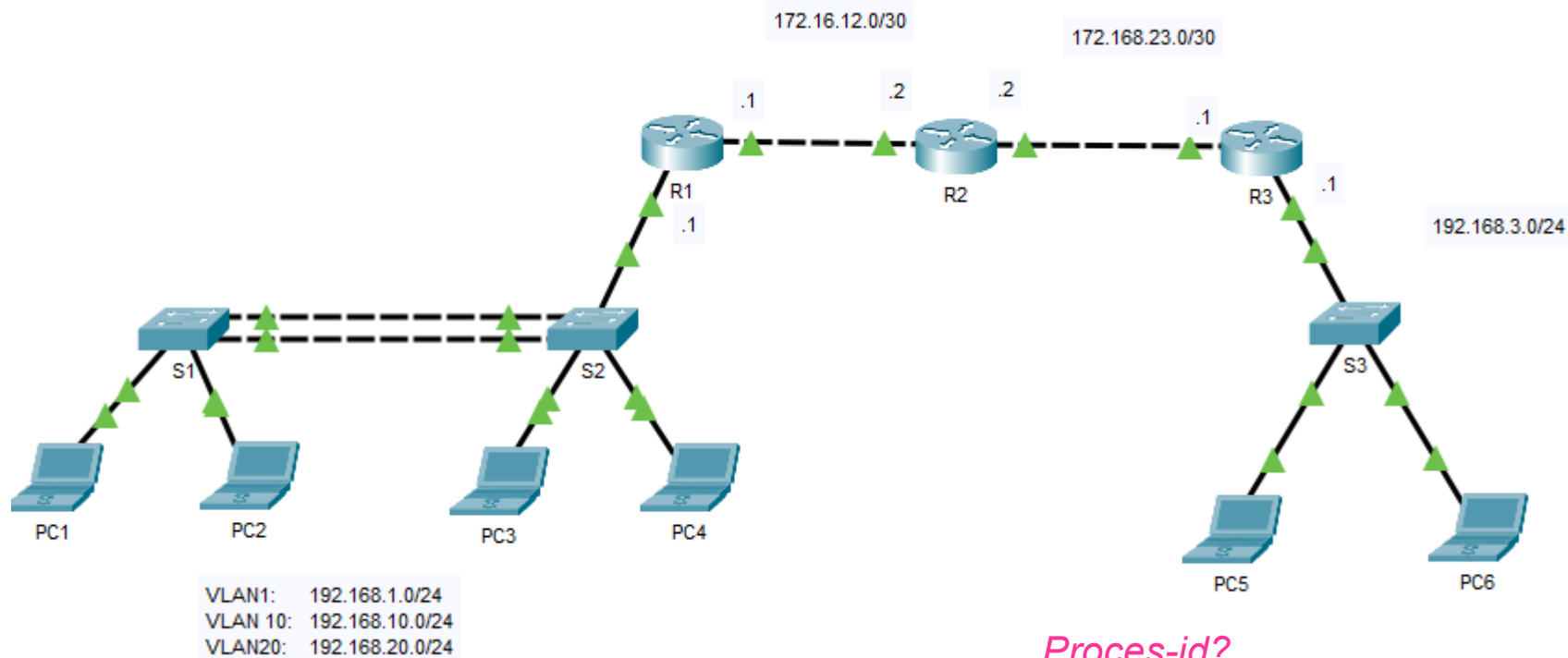




Meget basal konfiguration af OSPF

- På alle routerne i det område (area) der skal tale sammen, skal man aktivere og konfigurere OSPF.
- Konfigurationen går ud på at fortælle alle de andre routere, hvilke netværk "jeg" (altså den aktuelle router du er ved at konfigurere) er medlem af.
- Disse oplysninger sender OSPF så automatisk til de andre routere og alle har nu kendskab til alle netværk – også dem hos de andre routere 😊

Eksempel



Proces-id?
Wildcard mask?
Kort – vi kommer tilbage til det! 😊

R3(config)#router ospf 10

R3(config-router)# network 192.168.3.0 0.0.0.255 area 0

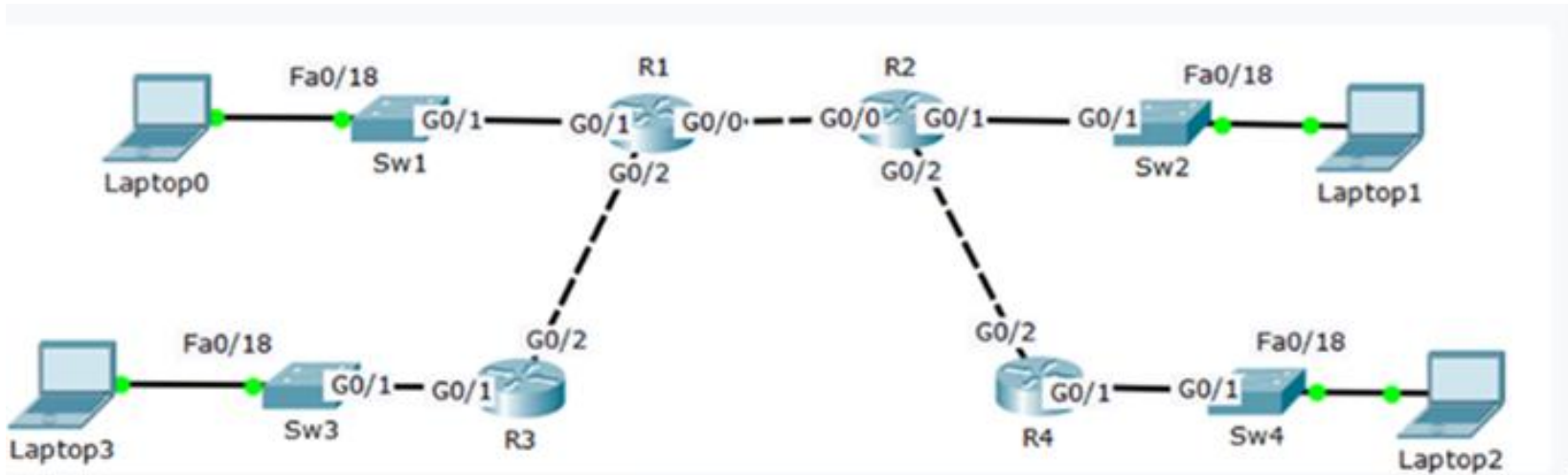
R3(config-router)# network 172.168.23.0 0.0.0.3 area 0

Hvordan med R1 og R2???? (fælles - på tavlen)

Ta-daaa... så nemt er det faktisk at konfigurere OSPF 😊



Byg dette netværk



- Du har netværket 10.10.0.0/16 til rådighed.
 - Dette netværk skal du opdele i 16 lige store netværk - nogle er til fremtidigt brug
1. Konfigurer OSPF single-area på routerne og kontrollér at de udveksler oplysninger (se i routingtabellen)
 2. Konfigurer passive interfaces på de interfaces der vender ind mod routerens LAN.
 3. Kontrollér at alle routere kender alle netværk i topologien – inklusiv de andre routers LAN.
 4. Kontrollér at alle PC'er kan ping hinanden

Husk for  at starte med at sætte labels på, **INDEN** du begynder at konfigurere noget som helst overhovedet!!!

- Løs alle punkter **BORTSET** fra punkt 2
- Og du skal ikke aflevere den.
- Pkt. 2 tager vi senere (og det tæller helt vildt **ned** hvis nogen spørger hvordan man gør det.... før vi har gennemgået det senere ☺)



OSPF “passive-interface”

```
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
```



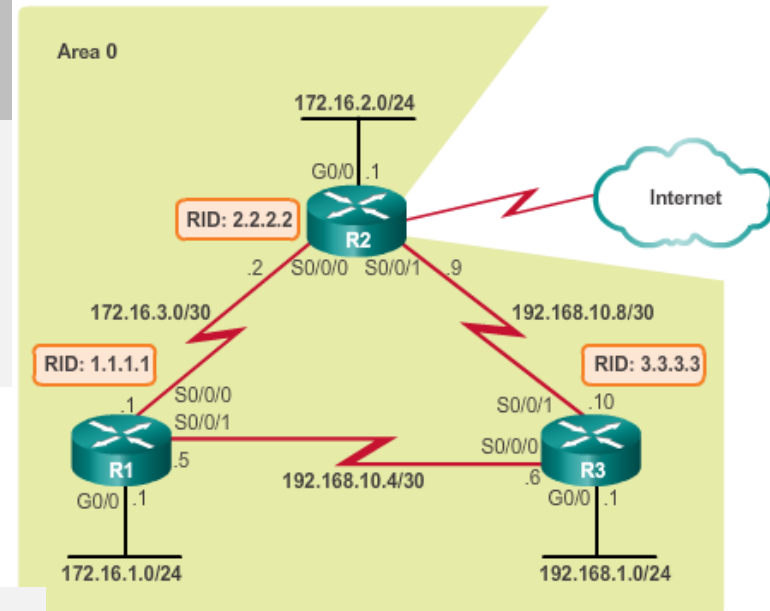
NB: Det er ikke altid der manuelt skal tildeles router-id. Hvis ikke du får at vide at du skal gøre det, så lader du bare routeren klare det selv 😊

```
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0
R1(config-router)# network 172.16.3.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.4 0.0.0.3 area 0
R1(config-router)#
R1(config-router)# passive-interface g0/0
R1(config-router)#
```

Nedenstående metode kan man også bruge – jeg er ikke vild med metoden 😊

```
R2(config)# router ospf 10
R2(config-router)# router-id 2.2.2.2
```

```
R2(config-router)# network 172.16.2.1 0.0.0.0 area 0
R2(config-router)# network 172.16.3.2 0.0.0.0 area 0
R2(config-router)# network 192.168.10.9 0.0.0.0 area 0
R2(config-router)#
R2(config-router)# passive-interface g0/0
R2(config-router)#
```



Hvad er "10-tallet" i #router ospf 10

????????????????????????????????????

Hvis nu vi ikke manuelt havde tastet OSPF router-id ind.....hvilken router-id ville R1, R2 og R3 så få?

Se svar på næste dias 😊



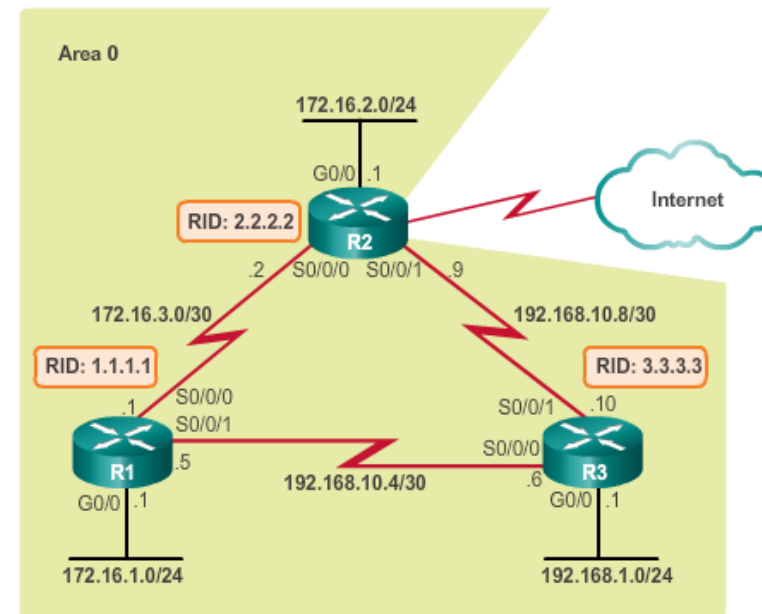
OSPF “router-id”

```
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
```

```
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0
R1(config-router)# network 172.16.3.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.4 0.0.0.3 area 0
R1(config-router)#
R1(config-router)# passive-interface g0/0
R1(config-router)#
```

```
R2(config)# router ospf 10
R2(config-router)# router-id 2.2.2.2
```

```
R2(config-router)# network 172.16.2.1 0.0.0.0 area 0
R2(config-router)# network 172.16.3.2 0.0.0.0 area 0
R2(config-router)# network 192.168.10.9 0.0.0.0 area 0
R2(config-router)#
R2(config-router)# passive-interface g0/0
R2(config-router)#
```



(#router ospf 10) 10 = process-id

Router-id (= HØJESTE IP adresse!)
Det kommer vi tilbage til om lidt!

R1: 192.168.10.5
R2: 192.168.10.9
R3: 192.168.10.10

*Hvad hvis der havde været
Loopback-interfaces?*



Verifying OSPF

```
R1# show ip ospf
Routing Process "ospf 10" with ID 1.1.1.1
Start time: 00:06:18.952, Time elapsed: 00:39:56.400

<Output omitted>

Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 1000 mbps
Area BACKBONE(0)
    Number of interfaces in this area is 3
Area has no authentication
```

```
R1# show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.1.0 0.0.0.255 area 0
    172.16.3.0 0.0.0.3 area 0
    192.168.10.4 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.3.3.3          110          00:12:14
    2.2.2.2          110          00:12:46
  Distance: (default is 110)
```

Se note!

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	0	FULL/ -	00:00:32	192.168.10.6	Serial0/0/1
2.2.2.2	0	FULL/ -	00:00:38	172.16.3.2	Serial0/0/0

Nonstop Forwarding (NSF) betyder at routeren kan fortsætte med at videresende IP pakker, selvom den proces der udregner og vedligeholder posterne i routing tabellen, midlertidigt stopper med at virke (ikke en del af dette pensum! ☺)

Verifying OSPF

```
R1# show ip ospf interface
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 172.16.1.1/24, Area 0, Attached via Network
Statement
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
        0              1          no            no            Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 172.16.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    oob-resync timeout 40
    No Hellos (Passive interface)
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet Address 192.168.10.5/30, Area 0, Attached via Network
Statement
  Process ID 10, Router ID 1.1.1.1, Network Type POINT_TO_POINT,
Cost: 647
```

<Output omitted>

```
R1# show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0	10	0	172.16.1.1/24	1	DR	0/0	
Se0/0/1	10	0	192.168.10.5/30	647	P2P	1/1	
Se0/0/0	10	0	172.16.3.1/30	647	P2P	1/1	

```
R1#
```



Hello pakker på kryds og tværs blandt alle routere????

Det er ikke så hensigtsmæssigt hvis alle routere sender alle beskeder om ændringer, til alle andre routere – som så fortæller alle routere om den nyhed de lige har modtager (som måske har ændret deres topologi tabel).

Derfor gør man noget lidt smartere ☺



OSPF Designated Router

I stedet for at routerne “flooder” deres oplysninger (ligesom en broadcast til alle routerne), har OSPF fundet en løsning, der ligner den metode som STP bruger 😊

OSPF arbejder med en “sjakformand” = én af routerne holder styr på hvad der sker, i forhold til opdateringer af netværkstopologien. Den hedder DR!

- The designated router (DR) is the solution to managing adjacencies and flooding of LSAs on a multiaccess network.
- The backup designated router (BDR) is elected in case the DR fails.
- All other non-DR and non-BDR routers become DROTHERs. DROTHERs only form adjacencies with the DR and BDR.
- DROTHERs only send their LSAs to the DR and BDR using the multicast address 224.0.0.6.
- DR uses the multicast address 224.0.0.5 to send LSAs to all other routers. DR only router flooding LSAs.
- DR/BDR Elections only necessary on multiaccess networks.



Default DR/BDR Election Process

- The router with the **highest interface priority** is elected as the DR.
- The router with the **second highest interface priority** is elected as the BDR.
- Priority can be configured between 0-255. (Priority of 0 = router cannot become the DR; 1 = default value)
- If interface **priorities are equal**, then the router with **highest router ID** is elected DR and second highest the BDR
- Three ways to determine **router ID**:
 - Router ID can be **manually configured**.
 - If not configured, the ID determined by the **highest loopback IP address**.
 - If no loopbacks, the ID is determined by the **highest active IPv4 address**.



DR/BDR Election Process

DR remains the DR until one of the following occurs:

- The DR fails.
- The OSPF process on the DR fails or is stopped.
- The multiaccess interface on the DR fails or is shutdown.

(Modsat STP hvor rootbridge skifter hvis en “bedre” switch kommer til 😊)

If the DR fails, the BDR is automatically promoted to DR.

- There is then a new BDR election and the DROTHER with the higher priority or router ID is elected as the new BDR.

(denne BDR beholder så til gengæld sin rolle også hvis den gamle DR kommer tilbage og kræver sin “ret” – hvis den er konf. sådan – til at få sin rolle tilbage)



The OSPF Priority

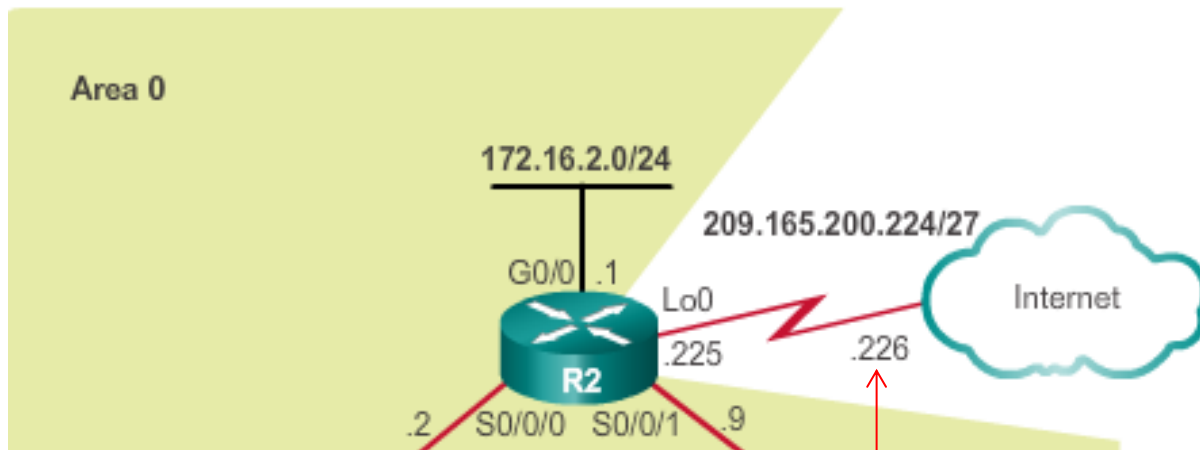
- Instead of setting the router ID on all routers, it is better to control the election by setting interface priorities. *Det er jeg ikke nødvendigvis enig i 😊*
 - To change the priority, use one of the following commands:
ip ospf priority value (OSPFv2 interface command)
- To begin another OSPF election, use one of the following methods:
 - Shutdown the router interfaces and then re-enable them starting with the DR, then the BDR, and then all other routers.
 - Reset the OSPF process using the **clear ip ospf process** privileged EXEC mode command on all routers.

```
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# ip ospf priority 255
R1(config-if)# end
R1#
```



Propagating a Default Static Route in OSPFv2

The router connected to the Internet that is used to propagate a default route is often called the edge, entrance or gateway router. In an OSPF network, it may also be call the **autonomous system boundary router (ASBR)**.



MEGET vigtig info 😊

```
R2(config)# ip route 0.0.0.0 0.0.0.0 209.165.200.226
R2(config)#
R2(config)# router ospf 10
R2(config-router)# default-information originate
R2(config-router)# end
R2#
```



OSPF Hello and Dead Intervals

OSPF Hello and Dead **intervals must match**, or a neighbor adjacency will not occur.

Verifying the OSPF Intervals on R1

```
R1# show ip ospf interface serial 0/0/0 | include Timer
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
R1#
```

Verifying OSPF Timer Activity

```
R1# show ip ospf neighbor
```

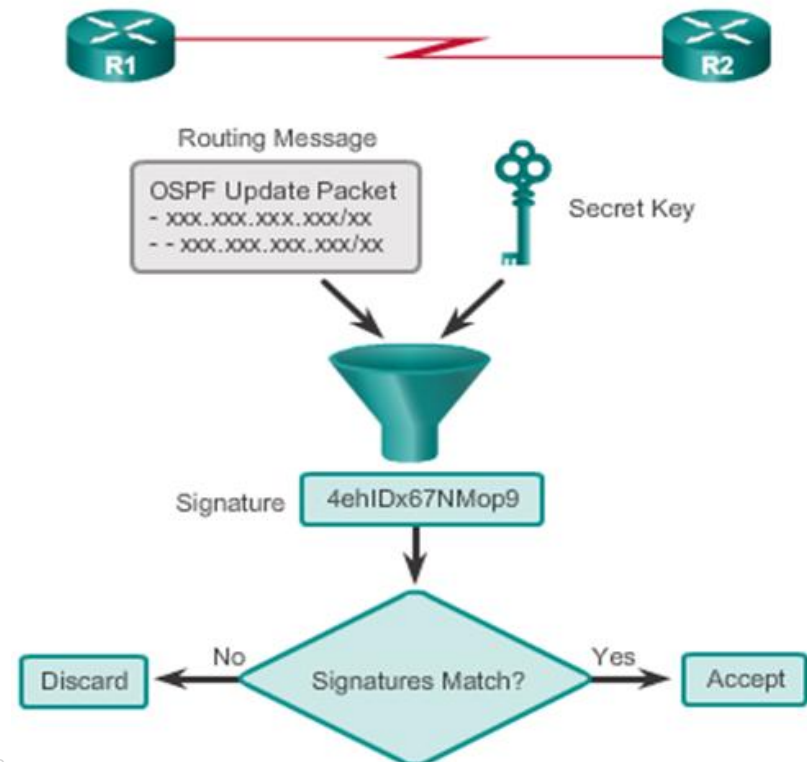
Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	0	FULL/-	00:00:35	192.168.10.6	Serial0/0/1
2.2.2.2	0	FULL/-	00:00:33	172.16.3.2	Serial0/0/0

```
R1#
```



Secure Routing Updates = password med i update pakker!

- When **neighbor authentication** has been configured on a router, the router authenticates the source of each routing update packet that it receives.
- OSPF supports three types of authentication:
 - Null** – no authentication.
 - Simple password authentication** – the password in the update is **sent in plaintext** over the network (**outdated method**).
 - MD5 authentication** – Most secure and recommended method of authentication. Password is **calculated using the MD5** algorithm.



Configuring OSPF MD5 Authentication

- To enable OSPF MD5 authentication **globally**

MD5 authentication can be enabled **globally for all interfaces or on a per-interface** basis (næsten det samme der skal gøres!)

```
R1(config)# router ospf 10
R1(config-router)# area 0 authentication message-digest
R1(config-router)# exit
R1(config)#
*Apr  8 09:58:09.899: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2
on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer
expired
R1(config)#
*Apr  8 09:58:28.627: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3
on Serial0/0/1 from FULL to DOWN, Neighbor Down: Dead timer
expired
R1(config)#
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# exit
R1(config)#
R1(config)# interface Serial 0/0/0
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# exit
R1(config)#
R1(config)# interface Serial 0/0/1
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# exit
```

```
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# ip ospf authentication message-digest
R1(config-if)# exit
R1(config)#
R1(config)# interface Serial 0/0/0
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# ip ospf authentication message-digest
R1(config-if)# exit
R1(config)#
R1(config)# interface Serial 0/0/1
R1(config-if)# ip ospf message-digest-key 1 md5 CISCO-123
R1(config-if)# ip ospf authentication message-digest
R1(config-if)# exit
R1(config)#
*Apr  8 10:20:10.647: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2
on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer
expired
R1(config)#
*Apr  8 10:20:50.007: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3
on Serial0/0/1 from FULL to DOWN, Neighbor Down: Dead timer
expired
R1(config)#
```

← To enable MD5 authentication on a **per-interface** basis



OSPF Troubleshooting Commands

- **show ip protocols** – Verifies vital OSPF configuration information.
- **show ip ospf neighbor** – Verifies that the router has formed an adjacency with its neighboring routers.
- **show ip ospf interface** – Displays the OSPF parameters configured on an interface, such as the OSPF process ID.
- **show ip ospf** – Examines the OSPF process ID and router ID.
- **show ip route ospf** – Displays only the OSPF learned routes in the routing table.
- **clear ip ospf [process-id] process** – Resets the OSPFv2 neighbor adjacencies.



Routing - fully specified route

A fully specified route refers to a static route entry in the [routing table](#) in which both the next-hop IP address and the exit interface are specified.

1.Next-Hop IP Address Only:

Here, you specify only the IP address of the next-hop router or device. No exit interface is indicated. The exit interface must be determined using **recursive routing**.

```
Router(config)# ip route 10.10.10.0 255.255.255.0 192.168.1.1
```

2.Exit Interface Only:

Here, you specify only the exit interface through which the packet should be sent. No next hop IP address is indicated.

```
Router(config)# ip route 10.10.10.0 255.255.255.0 GigabitEthernet 0/1
```



Routing - fully specified route

A fully specified route refers to a static route entry in the [routing table](#) in which both the next-hop IP address and the exit interface are specified.

3.Fully Specified (Both Next-Hop IP and Exit Interface):

Here, you specify both the IP address of the next-hop router/device and the exit interface.

```
Router(config)# ip route 10.10.10.0 255.255.255.0 192.168.1.1 GigabitEthernet 0/1
```

A fully specified route is often used in point-to-point connections, such as serial links, where there's only one possible destination over an interface. By specifying both the next-hop IP address and the exit interface, the router has precise information about where to forward the packet.

It's worth noting that while fully specifying a route can make routing determinate ("mere præcis") in certain scenarios, it can add to configuration complexity, so it's essential to use this method judiciously (velovervejet) and understand the design implications. *Så IKKE brug den, med mindre du har en god grund 😊*



OSPF øvelse i Learn. Ligner til forveksling øvelsen vi lige har lavet. Afleveres.

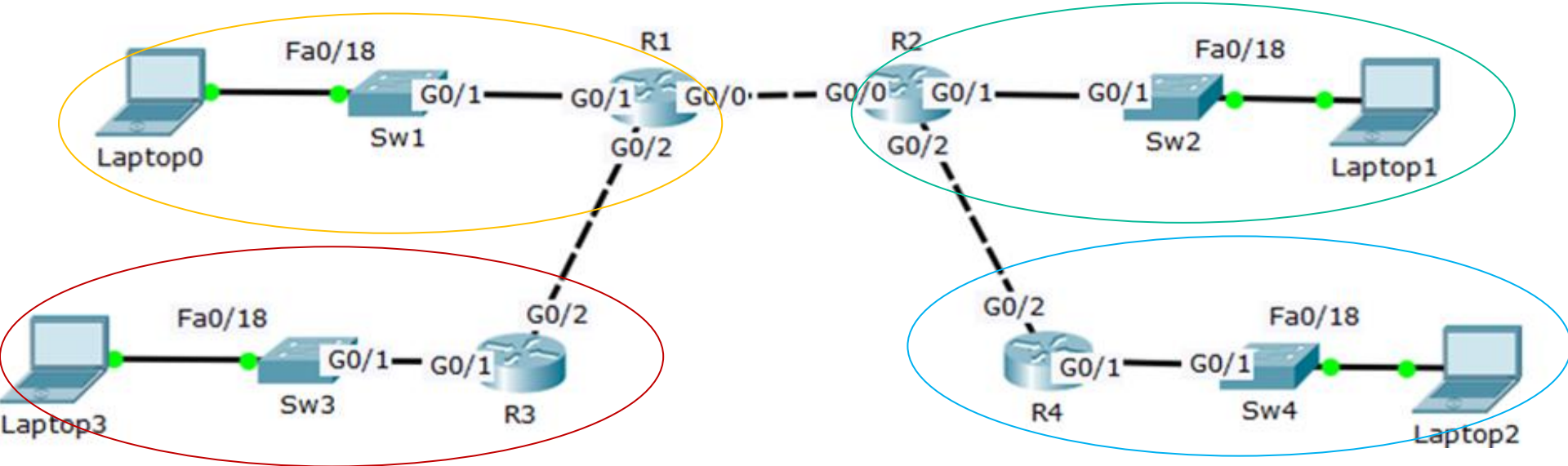
- Måske blev du ikke færdig i dag. Så fortsætter du i morgen – vigtigt at du får læst/set videoer



- Til i morgen, bedes du ENTEN at læse modul 1 og 2 i Netacad, ELLER at se de 6 videoer (ca. 1,5 timer) om OSPF på dette link:

- [CCNA 7-ENSA- Enterprise Networks, Security and Automation - YouTube](#)

1		CCNA 7 ENSA Enterprise Networks, Security and Automation- Module 1 Single Area OSPF Professor Awwad • 8.4K views • 3 years ago
2		CCNA 7 ENSA Enterprise Networks, Security and Automation- Module 2 OSPF Configuration part 1 Professor Awwad • 2.6K views • 3 years ago
3		CCNA 7 ENSA Enterprise Networks, Security and Automation- Module 2 OSPF Configuration part 2 Professor Awwad • 1.7K views • 3 years ago
4		CCNA 7 ENSA Enterprise Networks, Security and Automation- Module 2 OSPF Configuration part 3 Professor Awwad • 1.4K views • 3 years ago
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6		CCNA 7 ENSA Enterprise Networks, Security and Automation- Module 2 OSPF Configuration part 5 Professor Awwad • 947 views • 3 years ago



I må gerne hjælpe hinanden 2 og 2 – men alle skal selv bygge, på deres egen PacketTracer, på deres egen PC ☺ Nu skal I bygge denne opstilling ☺.
Hvor mange netværk er der her?

Du har netværket 192.168.200.0/24 til rådighed

Dette netværk skal opdeles (subnettes) i 8 undernetværk

1. Konfigurer OSPF single-area på routerne og kontrollér at de udveksler oplysninger (se i routingtabellen)
2. Konfigurer passive interfaces på de interfaces der vender ind mod routerens LAN.
3. Kontrollér at alle routere kender alle netværk i topologien – inklusiv de andre routeres LAN. Tag et skærmdump fra alle routere, som viser hvilke netværk de kender.
4. Kontrollér at alle PC'er kan ping hinanden
5. Aflever PT løsningen + skærmdumps i **Learn>Opgaver>Net2_OSPF single area**